

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · FORMULA BANK

## Chapter 6: We Distribute, Yet Things Multiply

Every formula and result of the chapter on one sheet

|    |                               |                                         |
|----|-------------------------------|-----------------------------------------|
| 1  | Distributive property         | $a(b + c) = ab + ac$                    |
| 2  | Product of two brackets       | $(a + m)(b + n) = ab + an + mb + mn$    |
| 3  | Square of a sum               | $(a + b)^2 = a^2 + 2ab + b^2$           |
| 4  | Square of a difference        | $(a - b)^2 = a^2 - 2ab + b^2$           |
| 5  | Difference of squares         | $(a + b)(a - b) = a^2 - b^2$            |
| 6  | Sum of the two squares        | $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$  |
| 7  | Difference of the two squares | $(a + b)^2 - (a - b)^2 = 4ab$           |
| 8  | Śrīdharācārya's square        | $a^2 = (a + b)(a - b) + b^2$            |
| 9  | Cube of a sum                 | $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ |
| 10 | Difference of cubes           | $(a - b)(a^2 + ab + b^2) = a^3 - b^3$   |
| 11 | Both numbers increased by 1   | $(a + 1)(b + 1) = ab + a + b + 1$       |
| 12 | Multiplying by 11             | $N \times 11 = 10N + N$                 |
| 13 | Multiplying by 99             | $N \times 99 = 100N - N$                |
| 14 | Consecutive numbers           | $n^2 - (n - 1)(n + 1) = 1$              |